

Standard Operating Procedure (SOP)

Krauss Maffei 1012-4300 GX WP – Injection Molding Machine

1. Purpose

To establish safe and standardized procedures for the operation of the Krauss Maffei 1012-4300 GX WP, ensuring operator safety and consistent part quality.

2. Scope

This SOP applies to all trained operators and technicians using the Krauss Maffei 1012-4300 GX WP at Integrity Tool & Mold Inc.

3. Responsibilities

- **Operators** are responsible for following this SOP and reporting any deviations or safety concerns.
 - **Leaders** must ensure only trained and certified personnel operate the machine.
 - **Maintenance Technicians** must ensure routine maintenance and safety checks are completed as scheduled.
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4. Required PPE (Personal Protective Equipment)

Operators must wear:

- Safety glasses
 - Heat-resistant gloves
 - Steel-toe safety boots
 - Long sleeves and pants (no synthetic materials near heated areas)
 - Hearing protection (encouraged)
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5. Pre-Operational Safety Checks

Before operating the machine:

- Perform a **360° visual inspection** of the machine.
 - Ensure **emergency stop buttons** are accessible and functional.
 - Inspect **hydraulic and electrical lines** for visible wear or leaks.
 - Check that all **guards and interlocks** are in place and not bypassed.
 - Confirm the **mold is properly mounted** and clamped securely.
 - Verify **cooling systems** and **ejection systems** are active and functional.
 - Review job setup sheet for material, mold, and parameter verification.
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6. Machine Start-Up Procedure

1. Power on the main electrical supply and control system.
 2. Check oil level, temperature, and hydraulics.
 3. Load required program and verify settings against the job setup sheet.
 4. Start barrel heating zones and wait until they reach set temperatures.
 5. Confirm the safety door interlocks are working before cycling the machine.
 6. Conduct a **test cycle** without injection to ensure mold movement and ejection work properly.
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7. Operating Procedure

1. Load material into the hopper (ensure correct resin type).
 2. Initiate the production cycle via the HMI (Human-Machine Interface).
 3. Closely monitor:
 - Clamp force
 - Injection pressure and speed
 - Holding time
 - Ejection system
 - Cooling time and part quality
 4. Ensure no one is within the guarded area while the machine is in operation.
 5. Stop production immediately if any unusual noise, odor, or mechanical behavior occurs.
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8. Shutdown Procedure

1. Finish the current cycle and remove the final molded part.
 2. Lower barrel temperature to standby or shutoff settings.
 3. Pull the wand and let the hopper suck up the remaining material in the hose, then shut the hopper loader off and run the remaining material out (as per shot size)
 4. Power down the machine via the control panel.
 5. Clean up the surrounding work area—dispose of purge material and scrap properly.
 6. Log production data and any issues in the machine logbook.
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9. Safety Protocols & Emergency Procedures

- **Emergency Stop:** Operators must be familiar with the E-stop locations.
 - **Lockout/Tagout (LOTO):** Must be used before performing maintenance, changing molds, or accessing guarded zones.
 - **High-Temperature Hazard:** Never touch barrel, nozzle, or mold without proper PPE.
 - **Hydraulic System:** Avoid standing directly under or near high-pressure areas during operation.
 - **Pinch Points:** Always be aware of moving platens and ejection systems.
 - **Material Handling:** Handle resins with gloves if required by MSDS.
 - **Incident Reporting:** Report all injuries, malfunctions, or unsafe conditions immediately.
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10. Training Requirements

All operators must complete:

- **SOP training** on the Krauss Maffei 1012-4300 GX WP
- **Lockout/Tagout (LOTO)** and **Emergency Response Training**
- **WHMIS** awareness for all resins used